

Aptamers: molecular tools for analytical applications.

PMID: 17581746 | Indexed in PubMed

Aptamers are artificial nucleic acid ligands, specifically generated against certain targets, such as amino acids, drugs, proteins or other molecules. In nature they exist as a nucleic acid based genetic regulatory element called a riboswitch. For generation of artificial ligands, they are isolated from combinatorial libraries of synthetic nucleic acid by exponential enrichment, via an in vitro iterative process of adsorption, recovery and reamplification known as systematic evolution of ligands by exponential enrichment (SELEX). Thanks to their unique characteristics and chemical structure, aptamers offer themselves as ideal candidates for use in analytical devices and techniques. Recent progress in the aptamer selection and incorporation of aptamers into molecular beacon structures will ensure the application of aptamers for functional and quantitative proteomics and high-throughput screening for drug discovery, as well as in various analytical applications. The properties of aptamers as well as recent developments in improved, time-efficient methods for their selection and stabilization are outlined. The use of these powerful molecular tools for analysis and the advantages they offer over existing affinity biocomponents are discussed. Finally the evolving use of aptamers in specific analytical applications such as chromatography, ELISA-type assays, biosensors and affinity PCR as well as current avenues of research and future perspectives conclude this review.

Aptasensors--the future of biosensing?

PMID: 11939212 | Indexed in PubMed

Aptamers are artificial nucleic acid ligands that can be generated against amino acids, drugs, proteins and other molecules. They are isolated from combinatorial libraries of synthetic nucleic acid by an iterative process of adsorption, recovery and reamplification. Aptamers, first reported in 1990, are attracting interest in the areas of therapeutics and diagnostics and offer themselves as ideal candidates for use as biocomponents in biosensors (aptasensors), possessing many advantages over state of the art affinity sensors. The properties of aptamers, their applicability to biosensor technology, current research and future prospects are addressed in this short review.

Aptamers: A Feasible Technology in Cancer Immunotherapy.

PMID: 27413756 | Indexed in PubMed

Aptamers are single-chained RNA or DNA oligonucleotides (ODNs) with three-dimensional folding structures which allow them to bind to their targets with high specificity. Aptamers normally show affinities comparable to or higher than that of antibodies. They are chemically synthesized and therefore less expensive to manufacture and produce. A variety of aptamers described to date have been shown to be reliable in modulating immune responses against cancer by either blocking or activating immune receptors. Some of them have been conjugated to other molecules to target the immune system and reduce off-target side effects. Despite the success of first-line treatments against cancer, the elevated number of relapsing cases and the tremendous side effects shown by the commonly used agents hinder conventional treatments against cancer. The advantages provided by aptamers could enhance the therapeutic index of a given strategy and therefore enhance the antitumor effect. Here we recapitulate

the provided benefits of aptamers with immunomodulatory activity described to date in cancer therapy and the benefits that aptamer-based immunotherapy could provide either alone or combined with first-line treatments in cancer therapy.

High-throughput and computational techniques for aptamer design.

PMID: 39390781 | Indexed in PubMed

Aptamers refer to short ssDNA/RNA sequences that target small molecules, proteins, or cells. Aptamers have significantly advanced diagnostic applications, including biosensors for detecting specific biomarkers, state-of-the-art imaging, and point-of-care technology. Molecular computation helps identify aptamers with high-binding affinity, enabling high-throughput screening, predicting 3D structures, optimizing aptamers for improved stability, specificity, and complex target interactions. Aptamers are versatile in the development of specific and sensitive diagnostics. However, there needs to be more understanding of the precise workflow that integrates sequence, structure, and interaction with the target. In this review, the author discusses how significant progress has been made in aptamer discovery using bioinformatics for sequence analysis, docking to model interactions, and MD simulations to account for dynamicity and predict free-energy. Furthermore, the author discusses how quantum chemical calculations are critical for modelling electronic structures and assigning spectroscopic signals. Incorporating machine learning into the aptamer discovery brings a transformative advancement. With NGS datasets, SELEX, and experimental structures, the implementation of newer workflows yields aptamers with improved binding affinity. Leveraging transfer learning to models using experimental structures and aptamer sequences expands the aptamer design space significantly. As ML continues to evolve, it is poised to become central in accelerating aptamer discovery for biomedical applications in the next 5 years.

Aptamers: A promising chemical antibody for cancer therapy.

PMID: 26863567 | Indexed in PubMed

Aptamers, also known as chemical antibodies, are single-stranded nucleic acid oligonucleotides which bind to their targets with high specificity and affinity. They are typically selected by repetitive in vitro process termed systematic evolution of ligands by exponential enrichment (SELEX). Owing to their excellent properties compared to conventional antibodies, notably their smaller physical size and lower immunogenicity and toxicity, aptamers have recently emerged as a new class of agents to deliver therapeutic drugs to cancer cells by targeting specific cancer-associated hallmarks. Aptamers can also be structurally modified to make them more flexible in order to conjugate other agents such as nano-materials and therapeutic RNA agents, thus extending their applications for cancer therapy. This review presents the current knowledge on the practical applications of aptamers in the treatment of a variety of cancers.

In vitro evolution of chemically-modified nucleic acid aptamers: Pros and cons, and comprehensive selection strategies.

PMID: 27715478 | Indexed in PubMed

Nucleic acid aptamers are single-stranded DNA or RNA oligonucleotide sequences that bind to a specific target molecule with high affinity and specificity through their ability to adopt 3-dimensional structure in solution. Aptamers have huge potential as targeted therapeutics, diagnostics, delivery agents and as biosensors. However, aptamers composed of natural nucleotide monomers are quickly degraded in vivo and show poor pharmacodynamic properties. To overcome this, chemically-modified nucleic acid aptamers are developed by incorporating modified nucleotides after or during the selection process by Systematic Evolution of Ligands by EXponential enrichment (SELEX). This review will discuss the development of chemically-modified aptamers and provide the pros and cons, and new insights on in vitro aptamer selection strategies by using chemically-modified nucleic acid libraries.

Aptamer-Functionalized DNA Nanostructures for Biological Applications.

PMID: 32030541 | Indexed in PubMed

DNA nanostructures hold great promise for various applications due to their remarkable properties, including programmable assembly, nanometric positional precision, and dynamic structural control. The past few decades have seen the development of various kinds of DNA nanostructures that can be employed as useful tools in fields such as chemistry, materials, biology, and medicine. Aptamers are short single-stranded nucleic acids that bind to specific targets with excellent selectivity and high affinity and play critical roles in molecular recognition. Recently, many attempts have been made to integrate aptamers with DNA nanostructures for a range of biological applications. This review starts with an introduction to the features of aptamer-functionalized DNA nanostructures. The discussion then focuses on recent progress (particularly during the last five years) in the applications of these nanostructures in areas such as biosensing, bioimaging, cancer therapy, and biophysics. Finally, challenges involved in the practical application of aptamer-functionalized DNA nanostructures are discussed, and perspectives on future directions for research into and applications of aptamer-functionalized DNA nanostructures are provided.

Preparation and characterization of aptamer-polyelectrolyte films and microcapsules for biosensing and delivery applications.

PMID: 26521977 | Indexed in PubMed

"Smart" materials are polymer systems that are able to change their physical or chemical properties in response to external stimuli in their environment. By adding a specific molecular recognition probe to a polymer, hybrid materials can be developed that retain the properties of the advanced polymer and gain the ability to respond to a specific molecular target. Aptamers are single-stranded oligonucleotides that are well-suited to serve as molecular recognition probes due to the specificity and affinity of their target recognition as well as their stability and ease of synthesis and labeling. In particular, their negatively charged backbone makes for their facile incorporation into polyelectrolyte-based materials. This article will provide a brief review of the currently reported biosensor and delivery platforms that have been reported employing aptamer-polyelectrolyte materials, as well as a detailed description of the methods used to synthesize and study films and microcapsules containing small-molecule aptamer probes.

Functional nucleic acid entrapment in sol-gel derived materials.

PMID: 24025165 | Indexed in PubMed

Functional nucleic acids (FNAs) are single-stranded DNA or RNA molecules, typically generated through in vitro selection, that have the ability to act as receptors for target molecules (aptamers) or perform catalysis of a chemical reaction (deoxyribozymes and ribozymes). Fluorescence-signaling aptamers and deoxyribozymes have recently emerged as promising biological recognition and signaling elements, although little has been done to evaluate their potential for solid-phase assays, particularly with species made of RNA due to their lack of chemical stability and susceptibility to nuclease attack. Herein, we present a detailed overview of the methods utilized for solid-phase immobilization of FNAs using a sol-gel entrapment method that can provide protection from nuclease degradation and impart long-term chemical stability to the FNA reporter systems, while maintaining their signaling capabilities. This article will also provide a brief review of the results of such entrapment studies involving fluorescence-signaling versions of a DNA aptamer, selected RNA-cleaving deoxyribozymes, and two different RNA aptamers in a series of sol-gel derived composites, ranging from highly polar silica to hydrophobic methylsilsesquioxane-based materials. Given the ability to produce sol-gel derived materials in a variety of configurations, particularly as thin film coatings on electrodes, optical fibers, and other devices, this entrapment method should provide a useful platform for numerous solid-phase FNA-based biosensing applications.

Selection and characterization of single stranded DNA aptamers for the hormone abscisic Acid.

PMID: 23971905 | Indexed in PubMed

The hormone abscisic acid (ABA) is a small molecule involved in pivotal physiological functions in higher plants. Recently, ABA has been also identified as an endogenous hormone in mammals, regulating different cell functions including inflammatory processes, stem cell expansion, insulin release, and glucose uptake. Aptamers are short, single-stranded (ss) oligonucleotides able to recognize target molecules with high affinity. The small size of the ABA molecule represented a challenge for aptamer development and the aim of this study was to develop specific anti-ABA DNA aptamers. Biotinylated abscisic acid (bio-ABA) was immobilized on streptavidin-coated magnetic beads. DNA aptamers against bio-ABA were selected with 7 iterative rounds of the systematic evolution of ligands by exponential enrichment method (SELEX), each round comprising incubation of the ABA-binding beads with the ssDNA sequences, DNA elution, electrophoresis, and polymerase chain reaction (PCR) amplification. The PCR product was cloned and sequenced. The binding affinity of several clones was determined using bio-ABA immobilized on streptavidin-coated plates. Aptamer 2 and aptamer 9 showed the highest binding affinity, with dissociation constants values of $0.98 \pm 0.14 \mu\text{M}$ and $0.80 \pm 0.07 \mu\text{M}$, respectively. Aptamers 2 and 9 were also able to bind free, unmodified ABA and to discriminate between different ABA enantiomers and isomers. Our findings indicate that ssDNA aptamers can selectively bind ABA and could be used for the development of ABA quantitation assays.